

SAMUEL AMOS-OSEBEYO

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Education

Imperial College London

Expected Graduation: May 2028

Electronic and Information Engineering: MEng (integrated Master's)

Research and Work Experience

Adaptive and Intelligent Robotics Lab

June 2026 – Sep 2026

Undergraduate Researcher

- Trained quadruped parkour policies in NVIDIA Isaac Lab with massively parallel GPU-accelerated PPO (rsl_rl), under EurekaVerse-style LLM-generated terrain curricula.
- Ported the damage-adaptive locomotion pipeline from Isaac Gym to Isaac Lab and rewrote the damage catalogue and training scheme, completing 20% more goals under simulated actuator and joint faults than the previous implementation.
- Deployed the damage-adaptive policy to a Unitree Go2, where it performs parkour from a single depth camera.

Dexory

June 2025 – Sep 2025

Junior Electronics Intern

- Designed, brought up and validated multi-layer sensor and telemetry PCBs with C++ firmware for warehouse robots.
- Built modular C++ firmware libraries for I2C communication and telemetry collection, raising sensor throughput by 30%.
- Ran system-level testing of electromechanical assemblies, characterising power draw and reducing bus latency.

Imperial Planetary Robotics Lab

Oct 2024 – May 2026

Junior Robotics Engineer

- Led the hardware redesign of a 6-DOF robotic arm for a Mars-rover platform, improving end-effector positioning accuracy, developing its embedded C++ firmware and multi-layer PCBs, and integrating it into the rover's ROS2 stack with MoveIt and a custom control node for smoother teleoperation.
- Contributed to a 16th-place global finish, 1st in the UK, out of 100+ teams in an international rover challenge.

Projects

Distributed CAN-Bus Joint Controller (in progress)

- Designing a motor-driver PCB that mounts on each joint of a robotic arm and daisy-chains power and CAN to the next joint, minimising wiring and making every joint an independently addressable actuator.
- Each board pairs an ESP32-S3 and TMC5160 driver for NEMA 17/23 steppers with an AS5047P absolute magnetic encoder on the joint output, allowing accurate joint control.

Franka Panda VR Teleoperation and Data Collection

- Engineered a low-latency VR teleoperation pipeline from scratch, driving a Franka Panda arm from a Meta Quest 3 headset.
- Architected a distributed controller with ZeroRPC and Polymetis for real-time control over libfranka, and built a PREEMPT_RT kernel from source with custom configuration and rebuilt NVIDIA drivers so teleoperation, control and data collection run on one machine.
- Collected over 100 demonstration trajectories in LeRobot format for VLA and HIL-SERL policy training.

Sample-Efficient RL for Goal-Conditioned Manipulation

- Implemented SAC with Hindsight Experience Replay from scratch on panda-gym pick-and-place (52% success), then extended building BC, demo-seeded SAC+HER and BC+RL fine tuning from first principles.
- Evaluated in and out of distribution: BC+RL and demo-seeded SAC+HER tied at 100% and matched on generalisation, with fine-tuning using half the environment steps and degrading far less than BC.

Autonomous Self-Balancing Registration Robot

- Led six engineers building a two-wheeled inverted-pendulum robot that explores an unknown venue and identifies people by face, and integrated all five subsystems against a fixed demo deadline.
- Co-developed the ROS2 localisation and mapping stack, fusing wheel odometry and gyroscopic yaw with LiDAR scan matching in an EKF, and replaced explore_lite with a custom frontier explorer after measuring 2s map-update latency.
- Owned the SCRFD and ArcFace face-recognition pipeline, matching embeddings by cosine similarity at over 80% accuracy.

Skills

Hardware and Electronics: PCB design (KiCad) | Embedded C++ | Motor drivers | CAN, SPI, I2C | Fusion 360

Software and Tools: C++ | C | Python | PyTorch | ROS2 | MoveIt | Isaac Lab | Isaac Gym | OpenCV | RT Linux | Git

Robotics and Learning: RL (PPO, SAC, HER) | Imitation learning | Sim-to-real | Teleoperation | SLAM | EKF